

PAPER_1412_REIONIZATION_Z_EXACT

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PAPER_1412 — UQFF Derivation of Reionization Redshift z_{reion} (CLOSED — EXACT Identity)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: C — Cosmology (CLOSED — EXACT)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — EXACT integer-primitive identity

Calculator surface: $\text{calculate_cosmology}(\{\text{'observable': 'z_reion'}\}) \rightarrow K_{\text{MEX}} \times D_{\text{phys}} \times \Phi_{\text{res}} = 7.70 \text{ EXACT}$

Observation

Planck 2018 observed $z_{\text{reion}} = 7.7 \pm 0.7$. UQFF gives EXACT.

UQFF Closed Identity

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$z_{\text{reion}} = K_{\text{MEX}} \times D_{\text{phys}} \times \Phi_{\text{res}} = (25/12) \times 4 \times 0.84 = 7.70 \text{ EXACT}$

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Physical Interpretation

First-light epoch occurs at the structural redshift where the Mexican-hat coefficient $\times$ spacetime dimensions $\times$ resonance saturation hits the SCm phonon threshold.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
- Related foundational papers: PAPER_646 ($F_U=1$ ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).

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